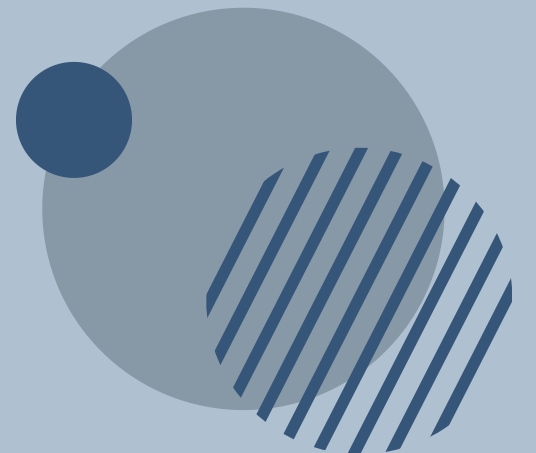




# CAPSTONE PROJECT

## AI-BASED ANIMAL-VEHICLE COLLISION DETECTION AND EARLY WARNING SYSTEM





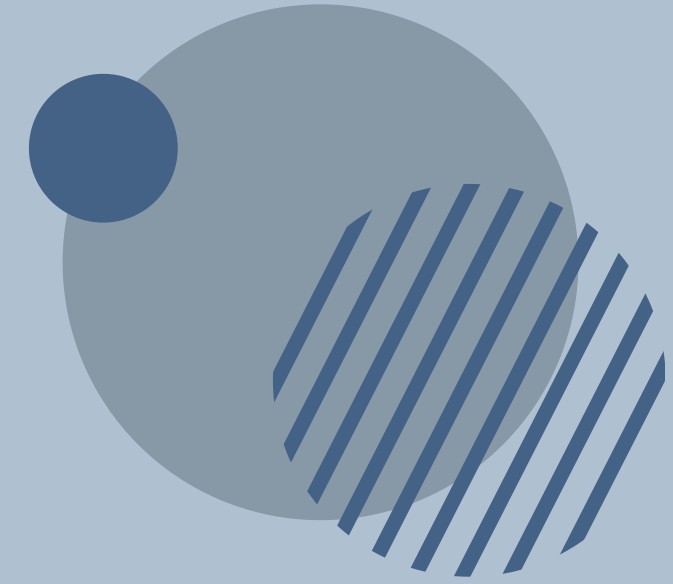
## INTRODUCTION

- Animal–vehicle collisions are a major cause of road accidents
- Common in highways, rural, and forest areas
- Sudden animal crossing leads to:
  - Vehicle damage
  - Injuries and fatalities
- Existing systems are passive and not real-time
- Need for an intelligent automated solution



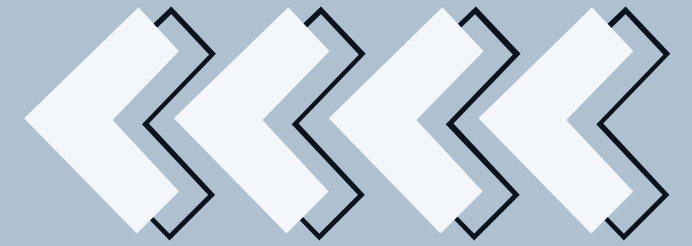


## PROBLEM STATEMENT

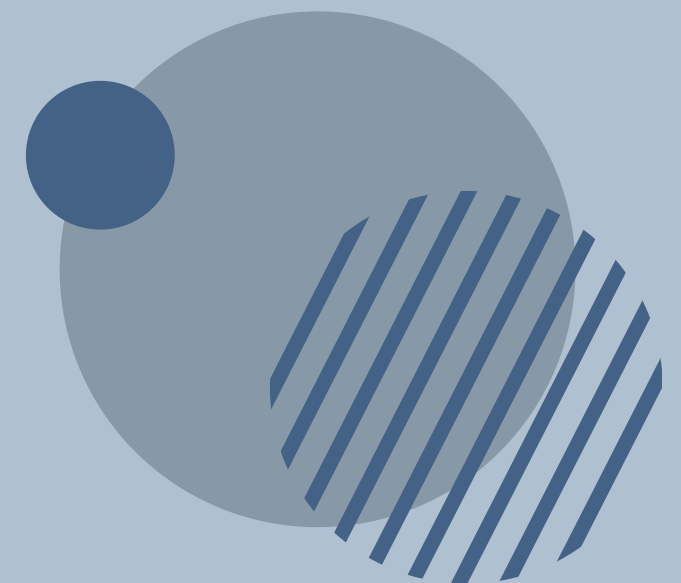
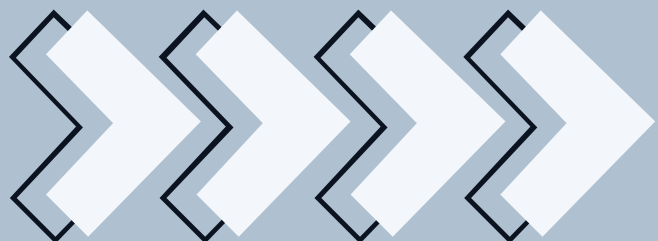
- No real-time detection in existing systems
  - Animals have:
    - Irregular movement
    - Different shapes and sizes
  - Traditional methods:
    - Road signs
    - GPS alerts
  - Limitations:
    - No live detection
    - No immediate warning
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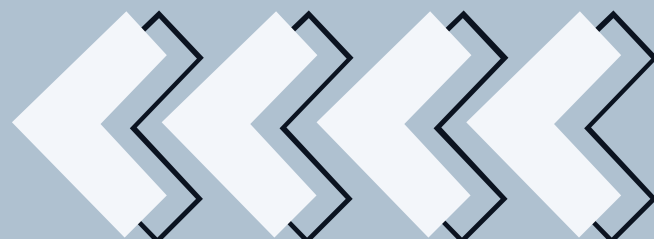


## PROPOSED SYSTEM

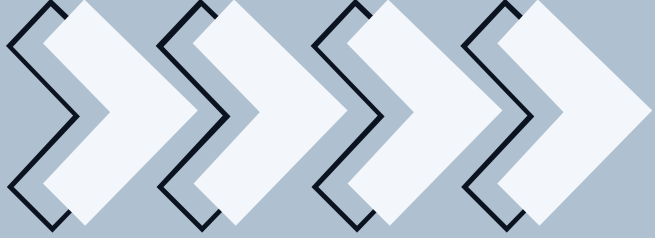
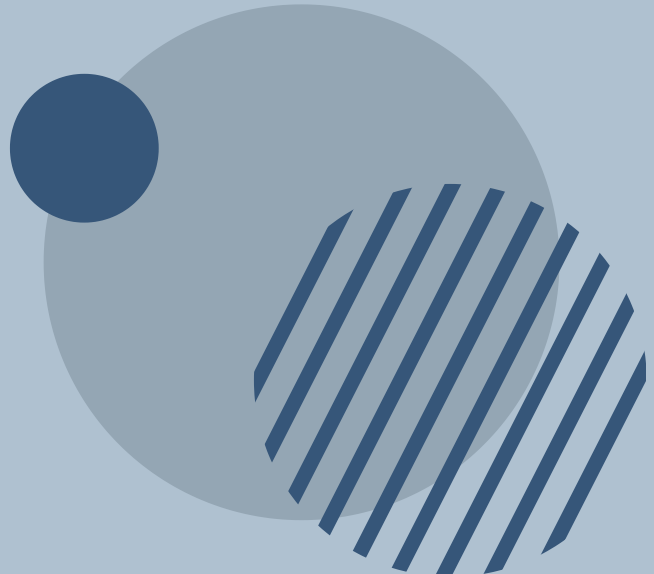


- AI-based real-time animal detection system
- Uses YOLOv8n deep learning model
- Input: Dashcam video
- Output:
  - Animal detection
  - Risk prediction
  - Driver alerts
- Works on CPU without GPU

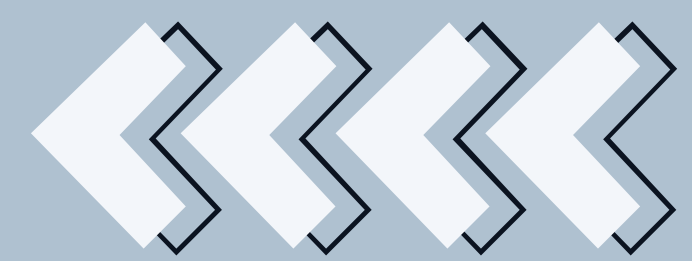




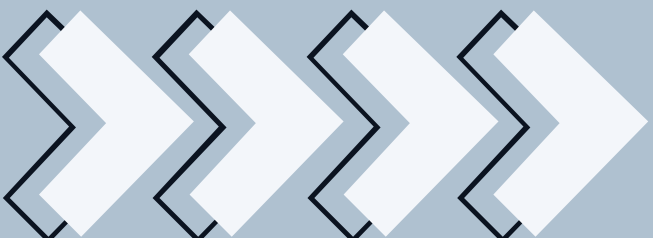
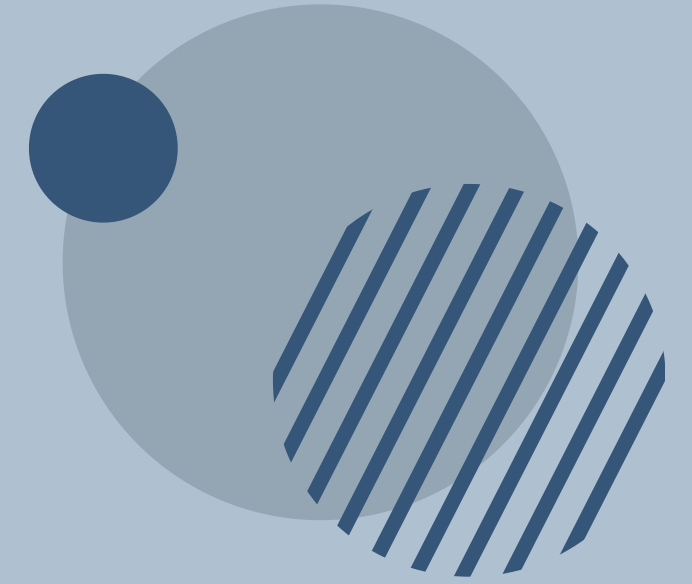
## OBJECTIVES

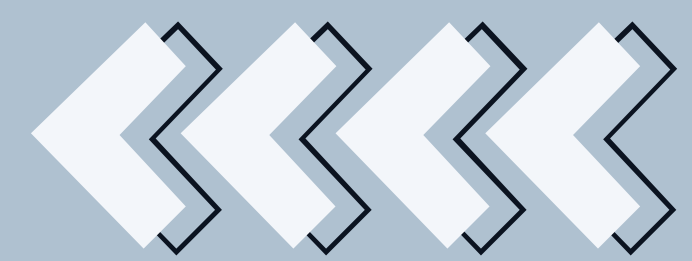
- Develop YOLOv8n-based detection model
  - Create unified dataset (single animal class)
  - Implement:
    - Real-time detection
    - Risk classification (LOW / MEDIUM / HIGH)
  - Provide:
    - Visual + audio alerts
  - Evaluate using:
    - Precision
    - Recall
    - mAP
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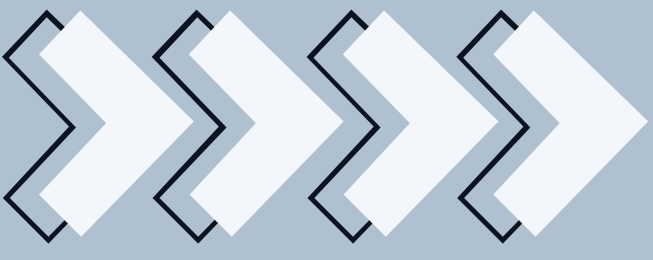
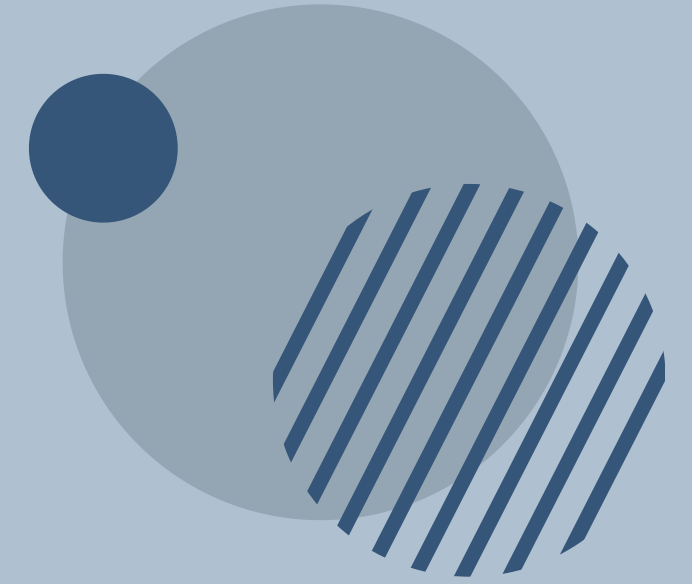


## SYSTEM FEATURES

- Real-time video processing
  - Bounding box detection
  - Risk classification
  - Collision prediction module
  - Dashboard display:
    - Distance
    - Direction
    - TTC
  - Automatic alert logging
  - Danger zone visualization
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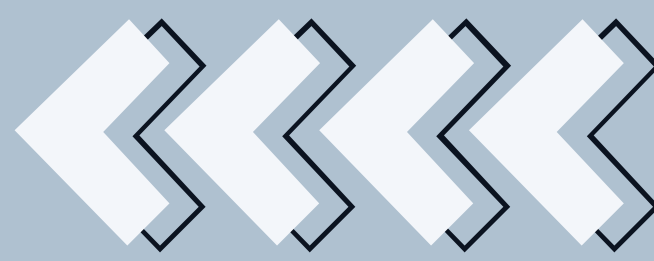
## EXISTING SYSTEM ANALYSIS

- Thermal cameras → Expensive
  - Radar systems → No classification
  - GPS alerts → Not real-time
  - Road signs → Passive
  - Older AI models → Slow or inaccurate
  - Need: Affordable + Real-time + Accurate system
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## DATASET DESCRIPTION

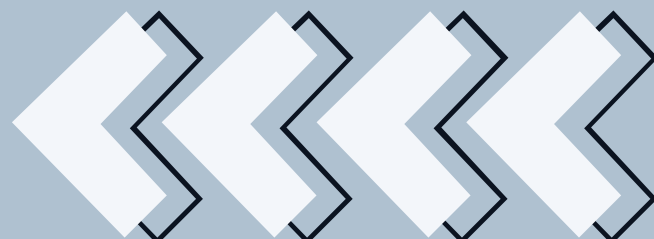
- Combined 3 Kaggle datasets:
  - Cows & buffalo
  - Cats & dogs
  - Wildlife animals
- Converted to single class: Animal
- Dataset stats:
  - Training images: 3,561
  - Validation images: 538
  - Resolution: 640×640



## DATA PREPROCESSING

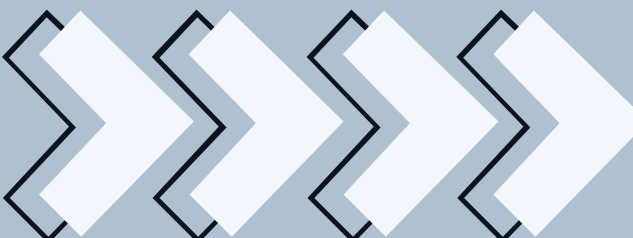
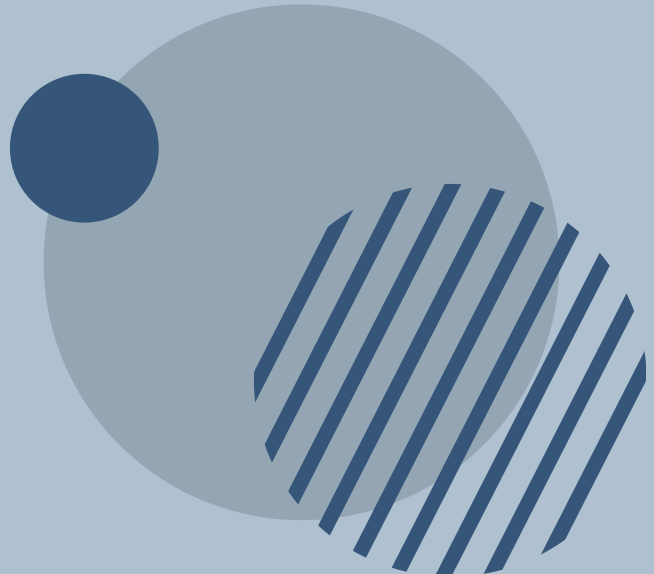
- Single-class conversion (all animals → class 0)
  - Removed:
    - Missing labels
    - Invalid images
  - Annotation validation
  - YAML configuration prepared
  - Ensured dataset consistency
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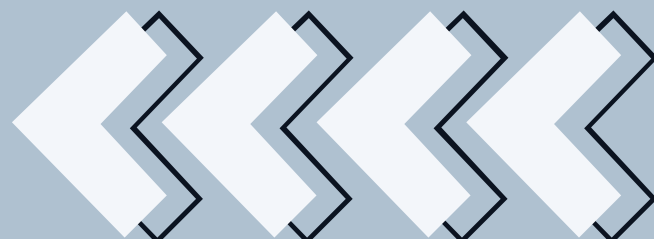
## METHODOLOGY

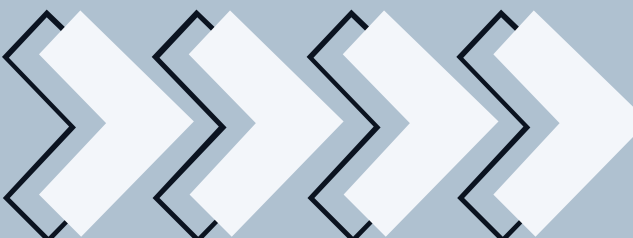
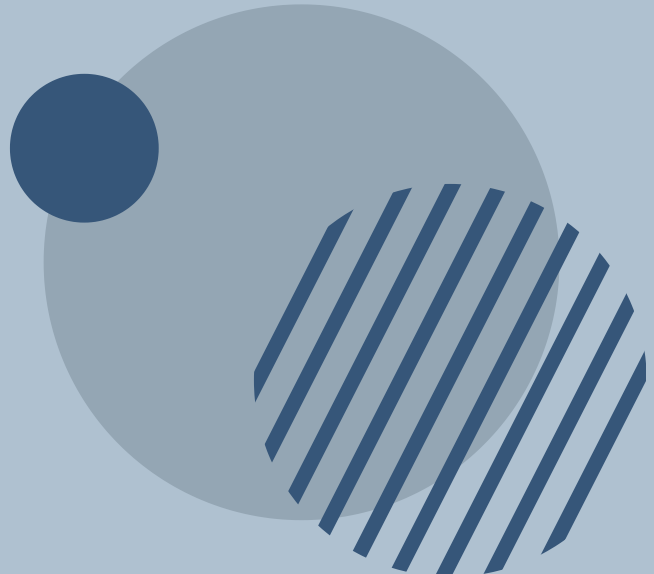
### Pipeline:

- Input video frames
  - YOLOv8 detection
  - Geometric filtering (false positives removal)
  - Temporal stability check
  - Risk classification
  - Collision prediction
  - Alert generation
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## MODEL BUILDING

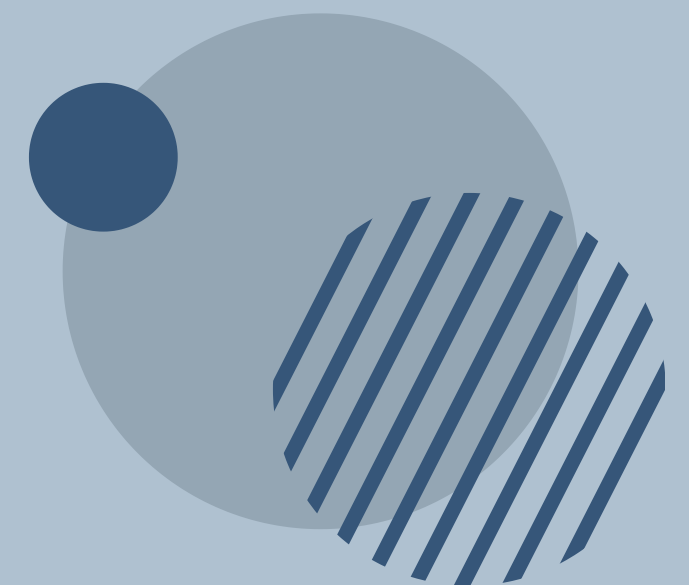
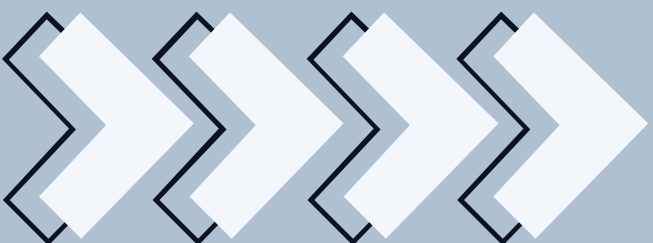
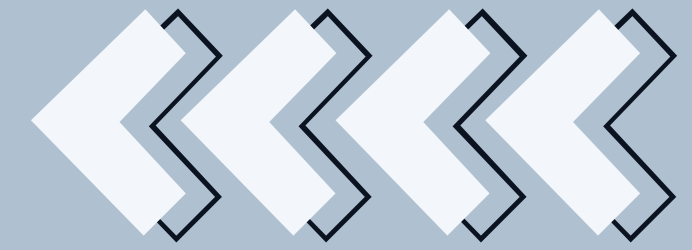


- Model: YOLOv8n
  - Training:
    - Epochs: 40
    - Image size: 640×640
    - Optimizer: AdamW
  - CPU-based training (~10.6 hours)
  - Rollback fine-tuning applied to improve accuracy
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## RISK CLASSIFICATION

- Based on:
  - Bounding box area ratio
  - Object position
  - Temporal stability
- Levels:
  - LOW → Far
  - MEDIUM → Moderate distance
  - HIGH → Very close
- Collision prediction:
  - SAFE
  - COLLISION POSSIBLE



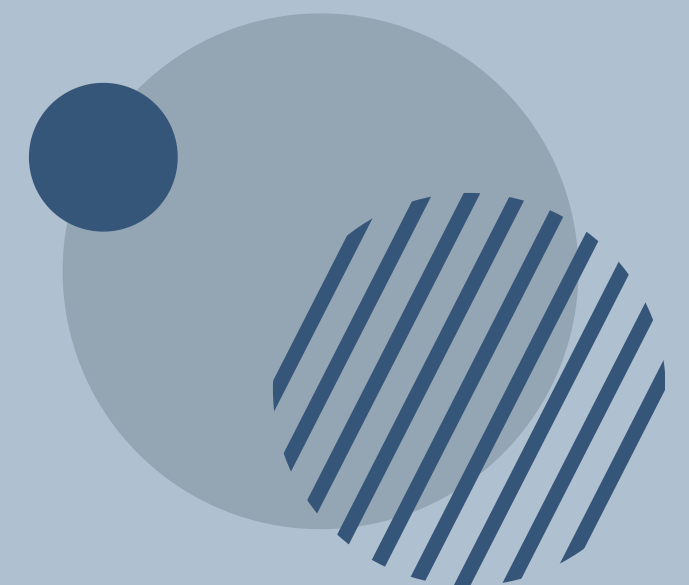
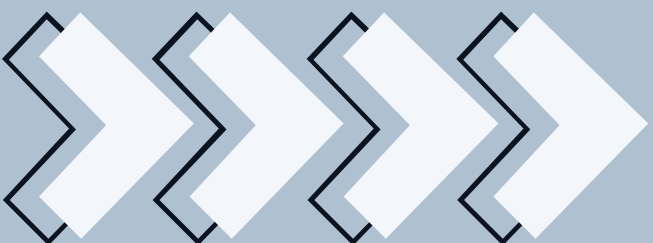


## PERFORMANCE METRICS

- Precision:  $\sim 0.82 - 0.84$
- Recall:  $\sim 0.79 - 0.80$
- mAP@50:  $\sim 83 - 84\%$
- mAP@50-95:  $\sim 58 - 59\%$

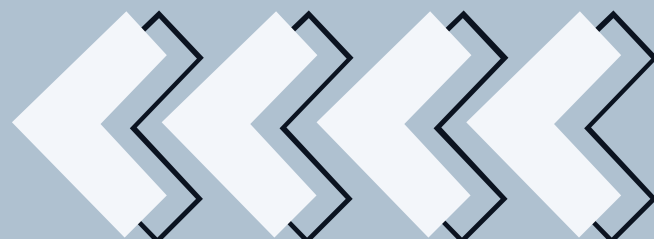
### Performance highlights:

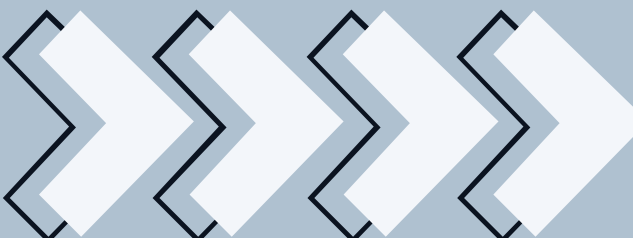
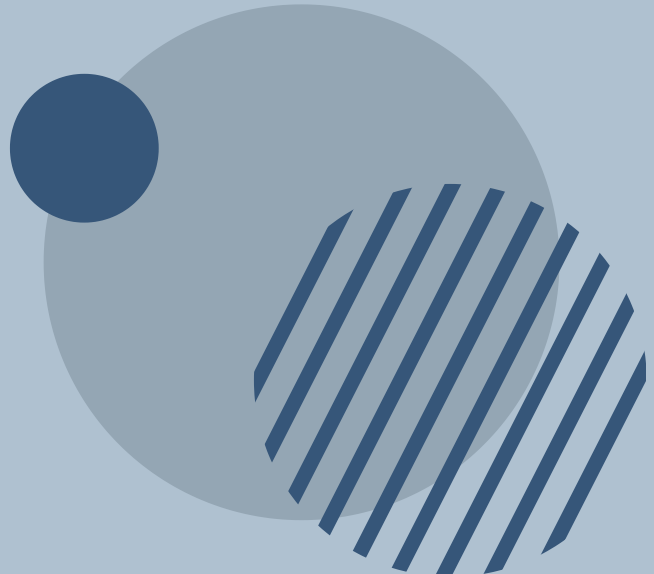
- Stable real-time detection on CPU
- Good balance between speed and accuracy
- Reduced false positives using filtering
- Consistent detection across different environments





## OUTPUT & ALERTS

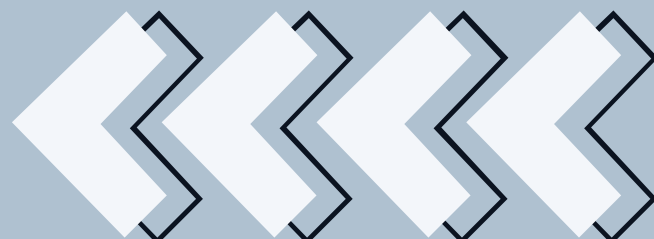


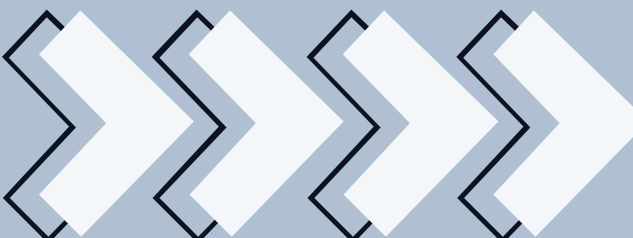
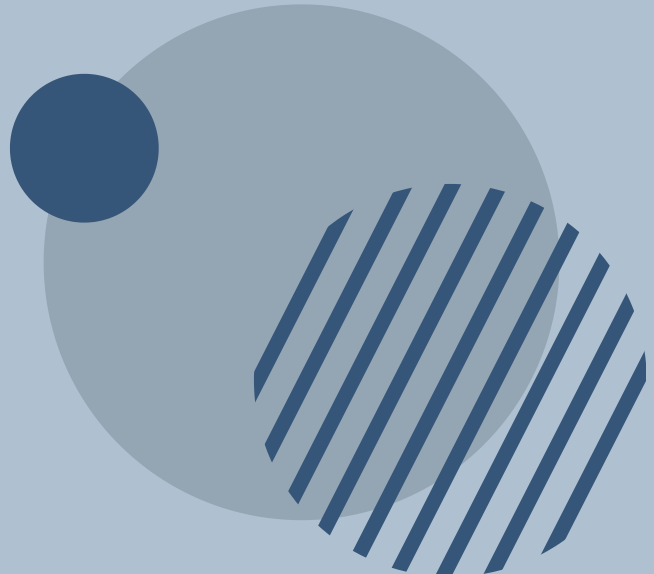
- Visual warning:
    - “ANIMAL AHEAD – SLOW DOWN”
  - Audio alert system
  - Dashboard shows:
    - Risk level
    - Distance
    - Direction
    - TTC
  - High-risk → Alert image saved automatically
- 
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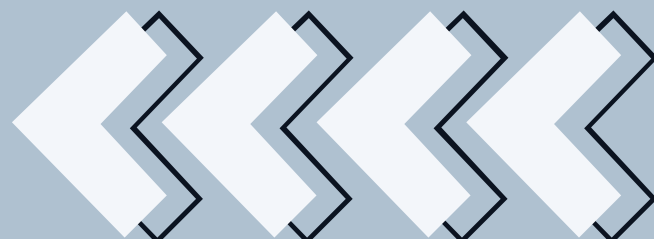




## BENEFITS



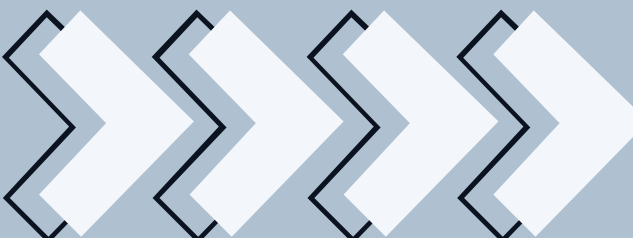
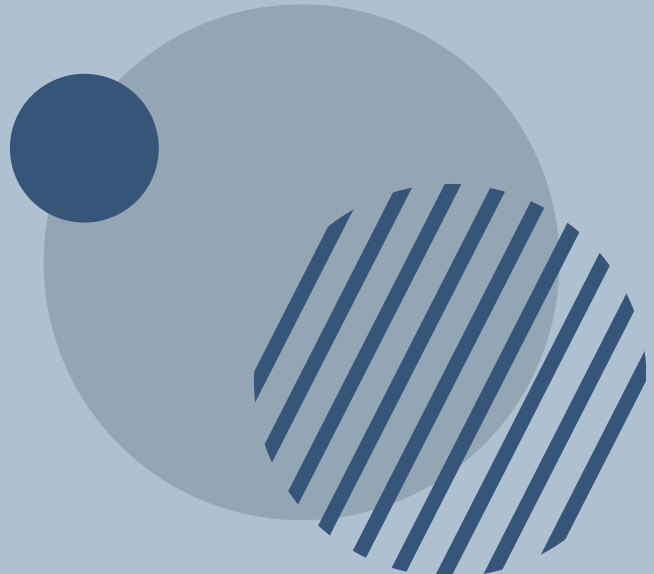
- Improves road safety
  - Reduces accidents
  - Protects wildlife
  - Saves economic loss
  - Works on low-cost hardware
  - Can be integrated into ADAS systems
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## CONCLUSION

- Developed a real-time AI-based detection system
- Achieved strong performance with YOLOv8n
- Efficient on CPU systems
- Provides early warning to drivers

Future scope:

- Multi-class detection
  - Sensor integration
  - Real-world deployment
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# THANK YOU



PRESENTED BY:  
ISHWARYA S

